

***Voice Card:*** My writing should sound professional, clear, engaging, technically accurate, confident, open to discussion where appropriate, and polite. Avoid unnecessary buzzwords or exaggerated claims.

***Before:*** This project demonstrates my passion for artificial intelligence and showcases my ability to build innovative solutions that solve real-world problems.

***After:*** This project began as an opportunity through my university internship rather than a problem I set out looking for. However, as I got deeper into development, the scope of the problem became clearer — I realized there was a genuine gap in how universities present information to prospective students, and my work naturally narrowed to focus on solving that specific pain point.

## **AI-Powered University Voice Assistant**

**An AI-powered university assistant that uses retrieval-augmented generation (RAG) and voice interaction to aid prospective and current students find university information quickly and accurately.**

### **Problem Statement**

University websites are often difficult to navigate, and each institution structures its information differently. For prospective students applying to multiple universities, this means repeating the same frustrating process — often spending 20 to 30 minutes simply searching for answers to simple questions (admission deadlines, program requirements, campus services) across varying site layouts. This project aims to remove that friction by building a voice assistant that can answer university-related questions instantly and naturally, regardless of how the source website is organized.

### **Reasoning**

This project began as an opportunity through my university internship rather than a problem I set out looking for. However, as I got deeper into development, the scope of the problem became clearer — I realized there was a genuine gap in how universities present information to prospective students, and my work naturally narrowed to focus on solving that specific pain point.

### **Key Decisions**

My approach was iterative, starting simple and increasing in complexity as requirements became clear:

- Prototyping first: I began by researching existing voice chatbots and evaluating available tools before committing to an architecture. I used LiveKit with initial prompt-based instructions to quickly test whether the core concept — a conversational voice bot — was viable.
- Moving to custom code: Once the prototype behaved as expected, I converted the LiveKit setup into a full codebase in VS Code, giving me control over the assistant's logic beyond what the low-code prototype allowed.

- Implementing RAG for accurate answers: To ground the assistant's responses in real university data rather than relying on general knowledge, I built a scraping pipeline to feed a Retrieval-Augmented Generation (RAG) system. This required multiple iterations (detailed below) as different scraping approaches failed on JavaScript-heavy sites.
- Testing in real conversational context: I validated the assistant's behaviour using LiveKit rooms, allowing me to test actual voice interactions rather than only text-based outputs.

## Technical and Design Trade-offs

At this stage, my focus has been entirely on the backend — UI/UX work is intentionally deferred to a later phase. Within the backend, I prioritized three things:

1. Response time — keeping the assistant fast enough to feel conversational rather than sluggish
2. Response accuracy — ensuring answers are grounded in real, scraped university data rather than hallucinated
3. Natural conversation flow — including custom-handled conversation endings, so interactions don't feel abrupt or robotic

These priorities meant I sometimes had to accept slower scraping or setup times to protect response quality once the assistant was live — a trade-off I considered worthwhile given that accuracy is the core value proposition for prospective students relying on this tool for real decisions.

## Challenges

The most significant challenge was building a reliable scraping pipeline for RAG:

- Simple scraping failed outright on most university sites, so I moved to alternative libraries for browser automation and rendering tools.
- Manual page-by-page scraping worked initially but wasn't scalable — university websites have far too many pages to add individually.
- When I tried to scrape site-wide automatically, I discovered that much of the content was JavaScript- and JSON-rendered, meaning standard scraping couldn't access it.
- I resolved this by combining HTTP requests, HTML rendering, and direct API fetches, which finally allowed consistent data extraction.
- After scraping was functional, I went back and improved the metadata layer to make retrieved content more useful for the RAG system.

This was a multi-week debugging process that required rethinking the approach several times rather than a single fix.

## Learning Curve

Beyond the technical skills — understanding how RAG systems work in practice and which scraping techniques hold up against modern, JavaScript-heavy websites — this project taught me patience. Solving the scraping problem took several failed attempts

before the right combination of tools worked. It also reminded me how enjoyable it is to experiment with ideas and iterate toward a solution, rather than expecting the first approach to work.

## **Outcome**

The project is currently in active development as part of my ongoing internship. At this stage, I have a functioning scraping and RAG pipeline that reliably retrieves and processes university website data, and the voice assistant is operational and testable through LiveKit rooms. Next steps include expanding the scope beyond backend functionality into UI/UX design.

## **About Me**

Starting my final year (7th semester) of my Computer Science degree in September, while simultaneously completing this AI voice assistant project as a university-sponsored internship. Alongside this, I'm interning at FlyRank AI, working on additional projects, and writing a research paper on how AI models predict the next token during generation under the mentorship of my professor. I'm also pursuing the AWS AI Practitioner certification and building foundational experience in project management.

## **Let's Connect**

I'm open to new opportunities, feedback, and collaboration — especially on projects involving applied AI, conversational systems, or research at the intersection of language models and real-world tools. If you'd like to discuss a potential project or just exchange ideas, feel free to reach out.

- **GitHub:** [github.com/AmnaMalik04](https://github.com/AmnaMalik04)
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